

ECON 320 — LECTURE NOTES

Comparative Statics and the Concept of Derivative

Based on Chiang & Wainwright (2005), Chapters 6–8

- Course** ECON 320: Introduction to Mathematical Economics
- Primary Text** A. C. Chiang & K. Wainwright, *Fundamental Methods of Mathematical Economics*, 4th ed., McGraw-Hill, 2005.
- Supplementary** Hoy et al., *Mathematics for Economics*, 3rd ed., MIT Press, 2011.
Simon & Blume, *Mathematics for Economists*, Norton, 1994.
- Topics** Comparative statics, limits, derivatives, differentiation rules,
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1 The Nature of Comparative Statics

1.1 What is Comparative Statics?

Comparative statics is one of the most fundamental tools of economic analysis. It asks: *if an exogenous parameter of a model changes, how does the endogenous (equilibrium) variable respond?* We compare two static equilibria — the “before” and “after” — without modelling the dynamic path between them.

The method permeates every field of economics:

- How does equilibrium price change when a tax is imposed?
- How does a firm’s optimal output change when input prices rise?
- How does the IS–LM equilibrium shift when the money supply expands?

Formally, an economic model typically specifies a system of equations in endogenous variables y and exogenous parameters α . Solving yields an equilibrium $y^* = f(\alpha)$. Comparative statics studies $dy^*/d\alpha$.

Definition — Comparative Statics

Comparative statics is the analysis of how the equilibrium value of an endogenous variable y^* changes in response to a change in an exogenous parameter α , holding all other exogenous variables fixed. The central object is the *comparative-static derivative*

$$\frac{dy^*}{d\alpha}$$

and its sign (direction) and magnitude (size).

1.2 A Simple Market Model

Consider the standard competitive market. Demand and supply are functions of price P and an exogenous shift parameter α (e.g. consumer income):

$$Q^d = D(P, \alpha), \quad Q^s = S(P)$$

Equilibrium requires $Q^d = Q^s$, i.e.

$$D(P^*, \alpha) - S(P^*) = 0 \tag{1}$$

The equilibrium price P^* is implicitly a function of α . To find $dP^*/d\alpha$, we differentiate (1) with respect to α (treating P^* as a function of α):

$$\frac{\partial D}{\partial P} \frac{dP^*}{d\alpha} + \frac{\partial D}{\partial \alpha} - \frac{dS}{dP} \frac{dP^*}{d\alpha} = 0$$

Solving:

$$\frac{dP^*}{d\alpha} = \frac{-\partial D/\partial \alpha}{\partial D/\partial P - \partial S/\partial P} \tag{2}$$

Worked Example — Market for a Normal Good

Let $D(P, \alpha) = a - bP + \alpha$ (linear demand, α = income) and $S(P) = c + dP$.

Equilibrium: $a - bP^* + \alpha = c + dP^*$, giving $P^* = \frac{a - c + \alpha}{b + d}$.

Direct differentiation:

$$\frac{dP^*}{d\alpha} = \frac{1}{b + d} > 0$$

since $b, d > 0$. A rise in consumer income shifts demand right and raises equilibrium price. Using formula (2): $\partial D / \partial \alpha = 1$, $\partial D / \partial P = -b$, $\partial S / \partial P = d$, so

$$\frac{dP^*}{d\alpha} = \frac{-1}{-b - d} = \frac{1}{b + d} > 0. \quad \checkmark$$

Comparative-static derivative for quantity: $Q^* = c + dP^* = c + \frac{d(a - c + \alpha)}{b + d}$,

so $\frac{dQ^*}{d\alpha} = \frac{d}{b + d} > 0$. Both price and quantity rise when income increases.

1.3 The Role of the Derivative

Comparative statics is therefore inseparable from the concept of the derivative. The derivative of $y^* = f(\alpha)$ measures the *instantaneous rate of change* of equilibrium with respect to the parameter — exactly what we need. The rest of this module builds the formal theory of derivatives that makes comparative-static analysis rigorous.

2 The Concept of Limit**2.1 Rates of Change and the Difference Quotient**

Before defining the derivative, we must understand the *rate of change* of a function over a finite interval and then let that interval shrink to zero.

Let $y = f(x)$. Suppose x changes from x_0 to $x_0 + \Delta x$ (where $\Delta x \neq 0$). The corresponding change in y is $\Delta y = f(x_0 + \Delta x) - f(x_0)$.

Definition — Difference Quotient

The **difference quotient** (or average rate of change) of f over the interval $[x_0, x_0 + \Delta x]$ is:

$$\frac{\Delta y}{\Delta x} = \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

Geometrically, this is the slope of the *secant line* through the points $(x_0, f(x_0))$ and $(x_0 + \Delta x, f(x_0 + \Delta x))$.

Worked Example — Difference Quotient for a Quadratic

Let $f(x) = x^2$. At $x_0 = 3$:

$$\frac{\Delta y}{\Delta x} = \frac{(3 + \Delta x)^2 - 3^2}{\Delta x} = \frac{9 + 6\Delta x + (\Delta x)^2 - 9}{\Delta x} = 6 + \Delta x$$

For $\Delta x = 2$: slope of secant = 8 (connecting (3, 9) and (5, 25)).

For $\Delta x = 1$: slope = 7. For $\Delta x = 0.1$: slope = 6.1.

As $\Delta x \rightarrow 0$, the slope approaches 6. This will be the derivative.

2.2 The Limit Concept**Definition — Limit (Informal)**

We say that $\lim_{x \rightarrow a} f(x) = L$ if $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a (but $x \neq a$). The value of f at a is irrelevant to the limit.

Definition — Limit (Formal — Epsilon-Delta)

$\lim_{x \rightarrow a} f(x) = L$ if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that:

$$0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon$$

Properties of Limits

If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$, then:

1. $\lim_{x \rightarrow a} [f(x) + g(x)] = L + M$
2. $\lim_{x \rightarrow a} [f(x) \cdot g(x)] = L \cdot M$
3. $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L}{M}$, provided $M \neq 0$
4. $\lim_{x \rightarrow a} [c \cdot f(x)] = c \cdot L$ for any constant c
5. $\lim_{x \rightarrow a} [f(x)]^n = L^n$ for positive integer n

These follow directly from the epsilon-delta definition (*Chiang & Wainwright*, p. 127; *Simon & Blume*, Ch. 2).

Worked Example — Computing Limits

1. $\lim_{x \rightarrow 2} (3x^2 - 5x + 1) = 3(4) - 5(2) + 1 = 3$ (polynomial, substitute directly)
2. $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$: substituting $x = 3$ gives $0/0$ (indeterminate). Factor: $\frac{x^2 - 9}{x - 3} = \frac{(x - 3)(x + 3)}{x - 3} = x + 3$ for $x \neq 3$. Therefore the limit = $3 + 3 = 6$.
3. $\lim_{\Delta x \rightarrow 0} \frac{(x + \Delta x)^2 - x^2}{\Delta x} = \lim_{\Delta x \rightarrow 0} (2x + \Delta x) = 2x$. (This is the derivative of

$$x^2.)$$

2.3 One-Sided Limits and Continuity

Definition — One-Sided Limits

The **right-hand limit** is $\lim_{x \rightarrow a^+} f(x) = L_R$ (approaching a from above).

The **left-hand limit** is $\lim_{x \rightarrow a^-} f(x) = L_L$ (approaching a from below).

The (two-sided) limit exists if and only if $L_R = L_L$.

Definition — Continuity

f is **continuous at** $x = a$ if:

- (i) $f(a)$ is defined,
- (ii) $\lim_{x \rightarrow a} f(x)$ exists,
- (iii) $\lim_{x \rightarrow a} f(x) = f(a)$.

f is **continuous on an interval** if it is continuous at every point of that interval.

Continuity Does Not Imply Differentiability

A function can be continuous everywhere but differentiable nowhere (Weierstrass function). More relevantly in economics: $f(x) = |x|$ is continuous at $x = 0$ but not differentiable there (the left- and right-hand derivatives differ). However, differentiability *does* imply continuity — see Theorem 3.2.

3 The Derivative

3.1 Definition of the Derivative

Definition — The Derivative

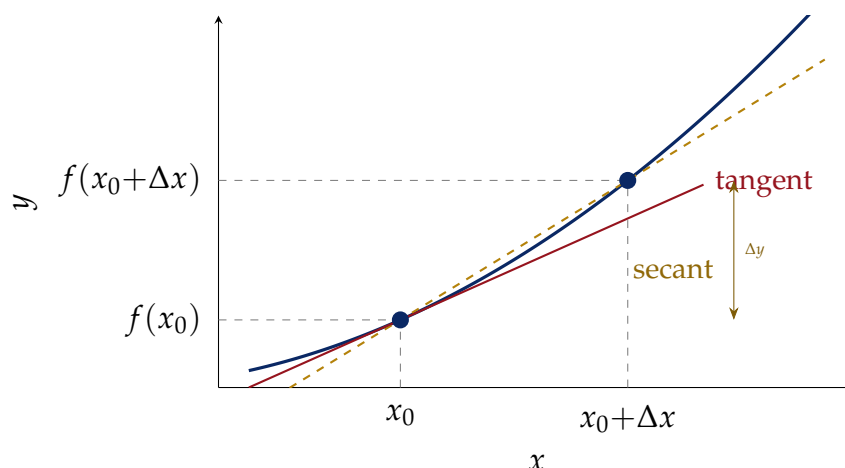
The **derivative** of f at x_0 , written $f'(x_0)$ or $\left. \frac{dy}{dx} \right|_{x=x_0}$, is defined as:

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

provided this limit exists. When it does, f is said to be **differentiable at** x_0 .

The **derivative function** $f' : \text{domain}(f) \rightarrow \mathbb{R}$ assigns to each x in the domain (where the limit exists) the value $f'(x)$.

Geometric interpretation. As $\Delta x \rightarrow 0$, the secant line through $(x_0, f(x_0))$ and $(x_0 + \Delta x, f(x_0 + \Delta x))$ rotates toward the *tangent line* at $(x_0, f(x_0))$. The derivative $f'(x_0)$ is the slope of this tangent line.



As $\Delta x \rightarrow 0$ the secant (gold) approaches the tangent (crimson); the derivative is the slope of the tangent.

3.2 Differentiability and Continuity

Theorem — Differentiability Implies Continuity

If f is differentiable at x_0 , then f is continuous at x_0 .

Proof. Since $f'(x_0)$ exists, write:

$$f(x_0 + h) - f(x_0) = \frac{f(x_0 + h) - f(x_0)}{h} \cdot h \xrightarrow{h \rightarrow 0} f'(x_0) \cdot 0 = 0$$

Therefore $\lim_{h \rightarrow 0} f(x_0 + h) = f(x_0)$, i.e. f is continuous at x_0 . $\square$

Worked Example — Deriving Derivatives from First Principles

1. $f(x) = c$ (constant function):

$$f'(x) = \lim_{h \rightarrow 0} \frac{c - c}{h} = \lim_{h \rightarrow 0} 0 = 0$$

2. $f(x) = x^n$ for $n \in \mathbb{N}$:

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h}$$

By the Binomial Theorem: $(x+h)^n = x^n + nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \dots + h^n$

$$f'(x) = \lim_{h \rightarrow 0} \frac{nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \dots}{h} = \lim_{h \rightarrow 0} \left[nx^{n-1} + \binom{n}{2}x^{n-2}h + \dots \right] = nx^{n-1}$$

3. $f(x) = \sqrt{x}$ for $x > 0$:

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} = \lim_{h \rightarrow 0} \frac{(\sqrt{x+h} - \sqrt{x})(\sqrt{x+h} + \sqrt{x})}{h(\sqrt{x+h} + \sqrt{x})} = \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} = \frac{1}{2\sqrt{x}}$$

4. $f(x) = e^x$:

$$f'(x) = \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h} = e^x \lim_{h \rightarrow 0} \frac{e^h - 1}{h} = e^x \cdot 1 = e^x$$

(using the standard result $\lim_{h \rightarrow 0} (e^h - 1)/h = 1$).

3.3 Notation

There are several equivalent notations for the derivative (*Chiang & Wainwright*, p. 132):

$$f'(x) = \frac{dy}{dx} = \frac{df}{dx} = \dot{y} = Df(x) = y'$$

The Leibniz notation dy/dx is especially useful in economics because it reminds us of the units (change in y per unit change in x) and is convenient for the chain rule.

4 Rules of Differentiation

4.1 Basic Rules

The following rules allow us to differentiate any combination of elementary functions without returning to the first-principles limit definition. All are proved using limit properties.

Theorem — Basic Differentiation Rules

Let f and g be differentiable at x , and $c \in \mathbb{R}$ a constant.

1. **Constant Rule:** $\frac{d}{dx}[c] = 0$
2. **Power Rule:** $\frac{d}{dx}[x^n] = nx^{n-1}$ for any $n \in \mathbb{R}$
3. **Constant Multiple:** $\frac{d}{dx}[c f(x)] = c f'(x)$
4. **Sum/Difference:** $\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$
5. **Product Rule:** $\frac{d}{dx}[f(x) g(x)] = f'(x) g(x) + f(x) g'(x)$
6. **Quotient Rule:** $\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{f'(x) g(x) - f(x) g'(x)}{[g(x)]^2}, \quad g(x) \neq 0$
7. **Chain Rule:** If $y = f(u)$ and $u = g(x)$, then $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = f'(g(x)) \cdot g'(x)$

Worked Example — Applying the Basic Rules

1. **Power and Sum:** $f(x) = 4x^3 - 7x^2 + 2x - 9$

$$f'(x) = 4(3x^2) - 7(2x) + 2(1) - 0 = 12x^2 - 14x + 2$$

2. Product Rule: $f(x) = (2x^2 + 1)(x^3 - 3x)$

Let $u = 2x^2 + 1$, $v = x^3 - 3x$, so $u' = 4x$, $v' = 3x^2 - 3$.

$$f'(x) = (4x)(x^3 - 3x) + (2x^2 + 1)(3x^2 - 3) = 4x^4 - 12x^2 + 6x^4 - 6x^2 + 3x^2 - 3 = 10x^4 - 15x^2 - 3$$

3. Quotient Rule: $f(x) = \frac{x^2 + 1}{x - 1}$, $x \neq 1$

$$f'(x) = \frac{2x(x - 1) - (x^2 + 1)(1)}{(x - 1)^2} = \frac{2x^2 - 2x - x^2 - 1}{(x - 1)^2} = \frac{x^2 - 2x - 1}{(x - 1)^2}$$

4. Chain Rule: $f(x) = (3x^2 + 5)^7$

Let $u = 3x^2 + 5$, $y = u^7$. Then $dy/du = 7u^6$, $du/dx = 6x$.

$$f'(x) = 7(3x^2 + 5)^6 \cdot 6x = 42x(3x^2 + 5)^6$$

4.2 Derivatives of Exponential and Logarithmic Functions

Standard Results for Exp and Log

$$\frac{d}{dx}[e^x] = e^x$$

$$\frac{d}{dx}[e^{ax}] = a e^{ax} \quad (a \in \mathbb{R})$$

$$\frac{d}{dx}[a^x] = a^x \ln a \quad (a > 0, a \neq 1)$$

$$\frac{d}{dx}[\ln x] = \frac{1}{x} \quad (x > 0)$$

$$\frac{d}{dx}[\ln |x|] = \frac{1}{x} \quad (x \neq 0)$$

$$\frac{d}{dx}[\log_a x] = \frac{1}{x \ln a} \quad (a > 0, a \neq 1, x > 0)$$

Worked Example — Exponential and Log Derivatives in Economics

1. Continuous compounding. Value of an investment: $V(t) = V_0 e^{rt}$. Rate of growth:

$$V'(t) = V_0 r e^{rt} = r V(t)$$

The instantaneous rate of growth is $V'(t)/V(t) = r$ — the continuous interest rate.

2. Log-linear demand. $\ln Q = a - b \ln P \Rightarrow Q = e^a P^{-b}$. The price elasticity:

$$\varepsilon = \frac{d \ln Q}{d \ln P} = -b$$

This is why log-linear (log-log) regressions yield constant elasticity.

3. Chain rule with log: $f(x) = \ln(3x^2 + 1)$. Let $u = 3x^2 + 1$:

$$f'(x) = \frac{1}{u} \cdot 6x = \frac{6x}{3x^2 + 1}$$

4.3 The Chain Rule and Composite Functions

The chain rule is indispensable in comparative statics, where equilibrium values depend on parameters through multiple layers of functional composition.

Chain Rule — Multiple Layers

If $y = f(u)$, $u = g(v)$, $v = h(x)$, then:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx} = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x)$$

Worked Example — Chain Rule in a Macro Model

In an IS-LM model suppose output Y depends on investment I , investment depends on interest rate r , and the interest rate depends on money supply M :

$$Y = f(I), \quad I = g(r), \quad r = h(M)$$

The multiplier dY/dM is:

$$\frac{dY}{dM} = \frac{dY}{dI} \cdot \frac{dI}{dr} \cdot \frac{dr}{dM} = f'(I) \cdot g'(r) \cdot h'(M)$$

If $f' > 0$ (higher investment raises output), $g' < 0$ (investment falls when interest rises), and $h' < 0$ (higher money supply lowers interest rate), then: $dY/dM = (+)(-)(-) > 0$: monetary expansion raises output.

4.4 The Inverse Function Rule

Theorem — Inverse Function Rule

If $y = f(x)$ is strictly monotone and differentiable with $f'(x) \neq 0$, then its inverse function $x = f^{-1}(y)$ is differentiable and:

$$\frac{dx}{dy} = \frac{1}{dy/dx} = \frac{1}{f'(x)}$$

Worked Example — Inverse Demand and Supply

Demand: $Q = 100 - 2P$. Then $\frac{dQ}{dP} = -2$.

Inverse demand: $P = 50 - Q/2$, so $\frac{dP}{dQ} = -\frac{1}{2}$.

Check: $\frac{dP}{dQ} = 1/(dQ/dP) = 1/(-2) = -1/2$. ✓

Application — Marginal Revenue: $TR = P \cdot Q = (50 - Q/2) \cdot Q = 50Q - Q^2/2$. $MR = d(TR)/dQ = 50 - Q$.

The slope of MR is exactly twice the slope of inverse demand. This is the standard result: $MR = P + Q (dP/dQ)$.

5 Higher-Order Derivatives

5.1 Second and Higher Derivatives

Definition — Higher-Order Derivatives

If $f'(x)$ is itself differentiable, its derivative is the **second derivative**:

$$f''(x) = \frac{d^2y}{dx^2} = \frac{d}{dx} \left[\frac{dy}{dx} \right] = D^2f(x) = y''$$

The n -th derivative is defined recursively: $f^{(n)}(x) = \frac{d}{dx}[f^{(n-1)}(x)]$.

Economic interpretations:

- If $y(t)$ is position at time t : $y'(t)$ is velocity, $y''(t)$ is acceleration.
- If $f(x)$ is a cost function: $f'(x) = MC$ (marginal cost), $f''(x)$ measures the rate of change of MC.
- If $f''(x) > 0$ at a critical point, we have a local minimum (second-order condition for minimisation).
- If $f''(x) < 0$ at a critical point, we have a local maximum (SOC for maximisation).

Worked Example — Higher Derivatives and Diminishing Returns

Production function: $Q = f(L) = 10L^{0.5}$ (square-root technology).

$$f'(L) = 5L^{-0.5} = \frac{5}{\sqrt{L}} > 0 \quad (\text{positive marginal product})$$

$$f''(L) = -\frac{5}{2}L^{-3/2} < 0 \quad (\text{diminishing marginal product})$$

The negative second derivative captures the law of diminishing returns: each additional unit of labour adds less to output than the previous unit.

Third derivative: $f'''(L) = \frac{15}{4}L^{-5/2} > 0$ — the rate of diminution is itself decreasing (marginal product falls at a decreasing rate).

5.2 Concavity, Convexity, and Inflection Points

Definition — Concavity and Convexity

Let f be twice differentiable on an interval I .

- f is **strictly concave** on I if $f''(x) < 0$ for all $x \in I$.
- f is **strictly convex** on I if $f''(x) > 0$ for all $x \in I$.

An **inflection point** is a point where f'' changes sign (concavity reverses); necessarily $f''(x_0) = 0$ there.

Worked Example — Cubic Cost Function

Total cost: $C(Q) = Q^3 - 6Q^2 + 15Q + 10$.

$$C'(Q) = 3Q^2 - 12Q + 15 \quad (\text{MC})$$

$$C''(Q) = 6Q - 12 \quad (\text{slope of MC})$$

$$C''(Q) = 0 \Rightarrow Q = 2.$$

For $Q < 2$: $C'' < 0$ (MC is falling — economies of scale).

For $Q > 2$: $C'' > 0$ (MC is rising — diseconomies of scale).

$Q = 2$ is the inflection point of C and the minimum of MC, corresponding to the *minimum efficient scale*.

6 Optimisation: First- and Second-Order Conditions

6.1 Unconstrained Optimisation in One Variable

Theorem — First- and Second-Order Conditions

Let f be twice differentiable on an open interval.

First-Order Condition (FOC): A necessary condition for x^* to be a local maximum or minimum is:

$$f'(x^*) = 0 \quad (\text{critical point})$$

Second-Order Condition (SOC):

- If $f'(x^*) = 0$ and $f''(x^*) < 0$: x^* is a **local maximum**.
- If $f'(x^*) = 0$ and $f''(x^*) > 0$: x^* is a **local minimum**.
- If $f'(x^*) = 0$ and $f''(x^*) = 0$: *inconclusive* — examine higher derivatives or use other methods.

Worked Example — Monopoly Profit Maximisation

Inverse demand: $P = 100 - Q$. Total cost: $C(Q) = Q^2 + 20Q + 50$.

Profit function: $\pi(Q) = (100 - Q)Q - (Q^2 + 20Q + 50) = -2Q^2 + 80Q - 50$

FOC:

$$\pi'(Q) = -4Q + 80 = 0 \Rightarrow Q^* = 20$$

SOC:

$$\pi''(Q) = -4 < 0 \quad \text{for all } Q$$

Confirms $Q^* = 20$ is a (global) maximum.

Equilibrium values: $P^* = 100 - 20 = 80$, $\pi^* = -2(400) + 80(20) - 50 = -800 + 1600 - 50 = 750$

Comparative statics: Suppose the cost function shifts — $C(Q) = Q^2 + 20Q + c$ where c is a fixed cost. Clearly Q^* does not depend on c (fixed costs don't affect the FOC). But if the variable cost shifts: $C(Q) = Q^2 + \alpha Q + 50$, then $\pi(Q) = -2Q^2 + (80 - \alpha)Q - 50$, FOC: $-4Q + (80 - \alpha) = 0 \Rightarrow Q^* = (80 - \alpha)/4$.

$$\frac{dQ^*}{d\alpha} = -\frac{1}{4} < 0$$

A rise in variable cost reduces monopoly output.

6.2 The Envelope Theorem

The Envelope Theorem provides an elegant shortcut for comparative statics: it tells us how the *optimal value function* changes with a parameter without re-solving the full optimisation.

Theorem — Envelope Theorem (One Variable)

Let $f(x; \alpha)$ be differentiable in both x and α . Suppose $x^*(\alpha) = \arg \max_x f(x; \alpha)$ is interior and satisfies the FOC $f_x(x^*; \alpha) = 0$. Define the value function $V(\alpha) = f(x^*(\alpha); \alpha)$. Then:

$$\frac{dV}{d\alpha} = \left. \frac{\partial f(x; \alpha)}{\partial \alpha} \right|_{x=x^*(\alpha)}$$

We differentiate f directly with respect to α , holding x fixed at x^* , ignoring the indirect effect through $x^*(\alpha)$.

Worked Example — Envelope Theorem in Producer Theory

A competitive firm maximises $\pi = Pf(L) - wL - rK$, where P (output price), w (wage), r (rental rate), and K are exogenous parameters.

FOC (choosing L): $Pf'(L^*) = w$.

By the Envelope Theorem:

$$\frac{d\pi^*}{dP} = \left. \frac{\partial \pi}{\partial P} \right|_{L=L^*} = f(L^*) = Q^* > 0 \quad (\text{higher price raises profit})$$

$$\frac{d\pi^*}{dw} = \left. \frac{\partial \pi}{\partial w} \right|_{L=L^*} = -L^* < 0 \quad (\text{higher wage lowers profit})$$

These are **Hotelling's Lemma**: $\partial \pi^* / \partial P = Q^*$ and $\partial \pi^* / \partial w = -L^*$.

7 Partial Derivatives

7.1 Definition and Interpretation

Most economic functions depend on more than one variable. A demand function $Q^d = D(P, P_s, P_c, Y)$ depends on own price, substitute price, complement price, and income. The partial derivative measures the effect of *one* variable while holding all others constant — the *ceteris paribus* condition, formalised.

Definition — Partial Derivative

For $z = f(x, y)$, the **partial derivative** with respect to x is:

$$\frac{\partial f}{\partial x} = f_x(x, y) = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}$$

obtained by differentiating with respect to x while treating y as a constant. Similarly, $\partial f / \partial y = f_y$ holds x fixed.

Worked Example — Computing Partial Derivatives

1. $f(x, y) = 3x^2y + 4xy^3 - 7y^2$

$$f_x = 6xy + 4y^3, \quad f_y = 3x^2 + 12xy^2 - 14y$$

2. **Cobb-Douglas:** $Q = AK^\alpha L^\beta$

$$\frac{\partial Q}{\partial K} = \alpha AK^{\alpha-1} L^\beta = \alpha \frac{Q}{K} \quad (\text{marginal product of capital})$$

$$\frac{\partial Q}{\partial L} = \beta AK^\alpha L^{\beta-1} = \beta \frac{Q}{L} \quad (\text{marginal product of labour})$$

3. **CES utility:** $U = (x_1^\rho + x_2^\rho)^{1/\rho}$

$$\frac{\partial U}{\partial x_1} = \frac{1}{\rho} (x_1^\rho + x_2^\rho)^{1/\rho-1} \cdot \rho x_1^{\rho-1} = \left(\frac{x_1^\rho + x_2^\rho}{x_1^\rho} \right)^{(1/\rho)-1} = U^{1-\rho} x_1^{\rho-1}$$

7.2 Second-Order Partial Derivatives

Definition — Second-Order Partial Derivatives

For $z = f(x, y)$, the four second-order partials are:

$$f_{xx} = \frac{\partial^2 f}{\partial x^2}, \quad f_{yy} = \frac{\partial^2 f}{\partial y^2}, \quad f_{xy} = \frac{\partial^2 f}{\partial x \partial y}, \quad f_{yx} = \frac{\partial^2 f}{\partial y \partial x}$$

f_{xy} and f_{yx} are **cross partial derivatives**.

Theorem — Young's Theorem (Symmetry of Cross Partial)

If f_{xy} and f_{yx} are both continuous at (x_0, y_0) , then:

$$f_{xy}(x_0, y_0) = f_{yx}(x_0, y_0)$$

The order of partial differentiation does not matter under continuity. (Chiang & Wainwright, p. 197; Simon & Blume, p. 336.)

Worked Example — Young's Theorem in Practice

$$f(x, y) = x^3y^2 + e^{xy}$$

$$f_x = 3x^2y^2 + ye^{xy}, \quad f_{xy} = 6x^2y + e^{xy} + xye^{xy} = 6x^2y + (1 + xy)e^{xy}$$

$$f_y = 2x^3y + xe^{xy}, \quad f_{yx} = 6x^2y + e^{xy} + xye^{xy} = 6x^2y + (1 + xy)e^{xy}$$

$$f_{xy} = f_{yx}. \quad \checkmark$$

7.3 The Total Differential**Definition — Total Differential**

For $z = f(x, y)$, the **total differential** is:

$$dz = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = f_x dx + f_y dy$$

It approximates the *total change* in z resulting from small changes dx and dy in both arguments simultaneously.

More generally, for $z = f(x_1, \dots, x_n)$:

$$dz = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i$$

Worked Example — Total Differential in Consumer Theory

Utility $U = f(x_1, x_2)$. Along an indifference curve, $dU = 0$:

$$dU = \frac{\partial U}{\partial x_1} dx_1 + \frac{\partial U}{\partial x_2} dx_2 = 0$$

$$\implies \frac{dx_2}{dx_1} = -\frac{\partial U / \partial x_1}{\partial U / \partial x_2} = -\frac{MU_1}{MU_2} = -MRS_{12}$$

The slope of the indifference curve equals the negative of the marginal rate of substitution. This derivation uses nothing beyond the total differential.

8 Implicit Function Theorem

8.1 Implicit Differentiation

Many economic relationships are defined implicitly — equilibrium conditions rather than explicit solved-out equations. We cannot always write $y = f(x)$ explicitly, but we can still find dy/dx by differentiating the implicit equation.

Theorem — Implicit Function Theorem (One Equation)

Let $F(y, x) = 0$ define y implicitly as a function of x near (y_0, x_0) . If F is continuously differentiable and $F_y \neq 0$ at (y_0, x_0) , then $y = y(x)$ exists locally and:

$$\frac{dy}{dx} = -\frac{F_x}{F_y}$$

(Chiang & Wainwright, p. 192; Hoyer et al., Ch. 14; Simon & Blume, Theorem 15.7.)

Derivation via Total Differential. Differentiate $F(y, x) = 0$ totally:

$$F_y dy + F_x dx = 0 \implies \frac{dy}{dx} = -\frac{F_x}{F_y} \quad (F_y \neq 0)$$

□

Worked Example — Market Equilibrium — Implicit Differentiation

Equilibrium condition: $F(P, t) = D(P) - S(P) - t = 0$

where t is an excise tax (the supply curve shifts by t).

$$F_P = D'(P) - S'(P), \quad F_t = -1$$

$$\frac{dP}{dt} = -\frac{F_t}{F_P} = -\frac{-1}{D'(P) - S'(P)} = \frac{1}{D'(P) - S'(P)}$$

Since $D' < 0$ (demand downward sloping) and $S' > 0$ (supply upward sloping):

$D'(P) - S'(P) < 0$, so $\frac{dP}{dt} < 0$?

Wait: note the sign. $D'(P) - S'(P) < 0$, so $dP/dt = 1/(\text{negative}) < 0$?

Recheck: an excise tax shifts the supply schedule up. Rewrite as $D(P) = S(P) + t$, so $F(P, t) = D(P) - S(P) - t = 0$. Tax up $\Rightarrow$ supply up $\Rightarrow$ price up.

We need: $D'(P) dP = S'(P) dP + dt \Rightarrow dP/dt = 1/(D' - S') > 0$?

Sign: $D' - S' = (\text{negative}) - (\text{positive}) < 0$, giving $dP/dt < 0$.

This contradicts intuition. The resolution: in the standard model we write the equilibrium as $D(P) = S(P - t)$ (tax on producers). Then $F = D(P) - S(P - t) = 0$, $F_P = D' - S'$, $F_t = S' > 0$, so $dP/dt = -S'/(D' - S') \in (0, 1)$ since $D' < 0 < S'$.

Result: Consumer price rises by $dP/dt = \frac{-S'}{D' - S'} = \frac{S'}{S' - D'} \in (0, 1)$.

Tax incidence on consumers: $dP/dt = S'/(S' - D')$; on producers: $1 - dP/dt = -D'/(S' - D')$. Both shares are positive and sum to 1.

Worked Example — Isoquant Slope via Implicit Differentiation

Production function $Q = F(K, L) = \bar{Q}$ defines an isoquant implicitly. Setting $F(K, L) - \bar{Q} = 0 \equiv G(K, L) = 0$:

$$\left. \frac{dL}{dK} \right|_{dQ=0} = -\frac{G_K}{G_L} = -\frac{F_K}{F_L} = -\frac{MP_K}{MP_L} = -\text{MRTS}_{KL}$$

The slope of the isoquant equals the negative of the Marginal Rate of Technical Substitution — derived purely from the implicit function rule.

8.2 Comparative Statics with Multiple Equations

When the model has n equations and n endogenous variables, matrix methods are needed. Differentiate the system $\mathbf{F}(\mathbf{y}; \alpha) = \mathbf{0}$ totally:

$$\mathbf{J} d\mathbf{y} + \mathbf{F}_\alpha d\alpha = \mathbf{0} \quad \Rightarrow \quad \frac{d\mathbf{y}}{d\alpha} = -\mathbf{J}^{-1} \mathbf{F}_\alpha$$

where $\mathbf{J}$ is the Jacobian matrix of partial derivatives.

Worked Example — IS-LM Comparative Statics

The IS-LM system in two endogenous variables (Y, r):

$$\text{IS: } Y - C(Y) - I(r) - G = 0$$

$$\text{LM: } L(Y, r) - \frac{M}{P} = 0$$

Total differentiation with respect to G (holding M/P fixed):

$$\begin{pmatrix} 1 - C'(Y) & -I'(r) \\ L_Y & L_r \end{pmatrix} \begin{pmatrix} dY \\ dr \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} dG$$

By Cramer's Rule (or matrix inversion), with $|J| = (1 - C')L_r - L_Y(-I')$:

$$\frac{dY}{dG} = \frac{L_r}{|J|}, \quad \frac{dr}{dG} = \frac{-(L_Y)}{|J|}$$

Standard sign assumptions: $0 < C' < 1$, $I' < 0$, $L_Y > 0$, $L_r < 0$.

$|J| = (1 - C')L_r + I'L_Y < 0 + (\text{negative}) < 0$.

$dY/dG = L_r/|J| = (\text{negative})/(\text{negative}) > 0$: fiscal expansion raises output (IS-LM multiplier).

9 The Concept of Elasticity

Definition — Point Elasticity

The **point elasticity** of y with respect to x measures the percentage change in y per one-percent change in x :

$$\varepsilon_{y,x} = \frac{dy/y}{dx/x} = \frac{dy}{dx} \cdot \frac{x}{y} = \frac{d \ln y}{d \ln x}$$

Elasticity is a *unit-free* measure of responsiveness.

Worked Example — Elasticities in Demand Analysis

1. **Linear demand:** $Q = 100 - 2P$.

$$\varepsilon_{Q,P} = \frac{dQ}{dP} \cdot \frac{P}{Q} = (-2) \cdot \frac{P}{100 - 2P}$$

At $P = 20$: $Q = 60$, $\varepsilon = (-2)(20/60) = -2/3$ (inelastic). At $P = 40$: $Q = 20$, $\varepsilon = (-2)(40/20) = -4$ (elastic).

2. **Constant-elasticity demand:** $Q = AP^{-b}$.

$$\varepsilon_{Q,P} = -bAP^{-b-1} \cdot \frac{P}{AP^{-b}} = -b \quad (\text{constant})$$

3. **Income elasticity of demand:** $Q = a - bP + cY$.

$$\varepsilon_{Q,Y} = c \cdot \frac{Y}{Q}$$

Normal good: $c > 0$ so $\varepsilon_{Q,Y} > 0$. Luxury: $\varepsilon > 1$.

4. **Output elasticity of capital (Cobb-Douglas):** $Q = AK^\alpha L^\beta$.

$$\varepsilon_{Q,K} = \frac{dQ}{dK} \cdot \frac{K}{Q} = \alpha AK^{\alpha-1} L^\beta \cdot \frac{K}{AK^\alpha L^\beta} = \alpha$$

The output elasticity equals the capital share parameter α .

10 Practice Questions

10.1 Section A: Limits and Derivatives from First Principles

A1. (First Principles) Use the definition $f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$ to find the derivative of each function at the indicated point.

(a) $f(x) = 3x^2 - 2x + 1$ at $x_0 = 2$

(b) $f(x) = 1/x$ at $x_0 = x$ (general point)

(c) $f(x) = \sqrt{x}$ at $x_0 = 4$

A2. (Limits) Evaluate each limit, showing all steps.

(a) $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$

(b) $\lim_{h \rightarrow 0} \frac{(2 + h)^3 - 8}{h}$

(c) $\lim_{x \rightarrow 0} \frac{e^{3x} - 1}{x}$

(d) $\lim_{x \rightarrow \infty} \frac{4x^2 - 3x}{7x^2 + 2}$

A3. (Continuity) Determine whether each function is continuous at the indicated point.

(a) $f(x) = \frac{x^2 - 4}{x - 2}$ at $x = 2$

(b) $g(x) = \begin{cases} x^2 & x < 1 \\ 2x - 1 & x \geq 1 \end{cases}$ at $x = 1$

(c) $h(x) = \frac{\sin x}{x}$ at $x = 0$ (define $h(0) = 1$)

10.2 Section B: Differentiation Rules

B1. (Standard Rules) Differentiate each function with respect to x .

(a) $f(x) = 5x^4 - 3x^3 + 2x^2 - 7x + 11$

(b) $g(x) = (2x^3 - 1)(x^2 + 3x)$

(c) $h(x) = \frac{x^2 - 2x + 1}{x^2 + 1}$

(d) $k(x) = e^{3x^2 - 1}$

(e) $\ell(x) = \ln\left(\frac{x^2 + 1}{x - 1}\right), x > 1$

(f) $m(x) = x^3 e^{-2x}$

B2. (Chain Rule) Differentiate using the chain rule.

(a) $y = (3x^2 + 2x - 1)^5$

(b) $y = e^{-x^2}$

(c) $y = \ln(\ln x), x > 1$

(d) $y = \left(\frac{x+1}{x-1}\right)^3, x \neq 1$

B3. (Inverse Function) Given $f(x) = 2x^3 + 3$, find df^{-1}/dy at the point where $y = 29$.

B4. (Higher-Order Derivatives) Find f' , f'' , and f''' for:

(a) $f(x) = x^5 - 4x^3 + 2x$

(b) $f(x) = xe^x$

(c) $f(x) = \ln x, x > 0$

10.3 Section C: Partial Derivatives and Total Differentials

C1. (Partial Derivatives) For each function, compute f_x, f_y, f_{xx}, f_{yy} , and f_{xy} .

(a) $f(x, y) = x^3y^2 - 4x^2y + 3y^3$

(b) $f(x, y) = e^{x^2+y^2}$

(c) $f(x, y) = \ln(xy) + x^2/y, x, y > 0$

C2. (Total Differential)

(a) Find dz for $z = x^2y^3 - 3xy + 5y$.

(b) For the production function $Q = K^{0.4}L^{0.6}$, find the total differential dQ and use it to approximate the change in output when K rises from 100 to 101 and L rises from 64 to 65.

C3. (Young's Theorem) Verify Young's Theorem for $f(x, y) = x^2e^y + 3x \ln y, y > 0$.

10.4 Section D: Comparative Statics and Implicit Differentiation

D1. (Implicit Differentiation)

(a) Find dy/dx if $x^3 + y^3 = 6xy$.

(b) Find dy/dx if $e^{xy} = x + y^2$.

(c) For the equilibrium condition $\alpha Q^2 - \beta Q + P = 0$ (with $\alpha, \beta > 0$), find dQ/dP using the implicit function rule.

D2. (Market Comparative Statics) A competitive market has:

$$Q^d = a - bP + cY, \quad Q^s = d + eP - fW$$

where Y is consumer income and W is the input wage.

(a) Find the equilibrium price P^* and quantity Q^* as functions of Y and W .

(b) Compute $dP^*/dY, dP^*/dW, dQ^*/dY, dQ^*/dW$.

(c) Provide the economic interpretation of each comparative-static result.

(d) How does the result change if $b + e = 0$? What does this imply economically?

D3. (Profit Maximisation — Comparative Statics) A monopolist faces demand $P = A - bQ$ and cost $C(Q) = cQ^2 + dQ + F$, where $A, b, c, d, F > 0$.

- (a) Derive the profit-maximising output Q^* and price P^* .
- (b) Use the envelope theorem or direct differentiation to find $dQ^*/dA, dQ^*/db, dQ^*/dc, dQ^*/dd$.
- (c) Show that $dP^*/dA \in (0, 1)$. What does this mean?
- (d) Find $d\pi^*/dF$ and $d\pi^*/dA$.

D4. (Elasticity)

- (a) Demand is $Q = 200 - 4P + 0.5Y$. At $P = 10, Y = 400$: (i) compute price elasticity and income elasticity; (ii) is demand elastic, inelastic, or unit-elastic at this point?
- (b) Show that for any demand function, $MR = P(1 + 1/\varepsilon_{Q,P})$.
- (c) For Cobb-Douglas $Q = AK^\alpha L^{1-\alpha}$ with $\alpha \in (0, 1)$, show that the output elasticity of each input equals its exponent, and that Euler's theorem implies constant returns to scale.

11 Full Solutions

11.1 Solutions: Section A

Solution

A1. Derivatives from First Principles

(a) $f(x) = 3x^2 - 2x + 1$, $x_0 = 2$:

$$\begin{aligned} f'(2) &= \lim_{h \rightarrow 0} \frac{3(2+h)^2 - 2(2+h) + 1 - [3(4) - 4 + 1]}{h} \\ &= \lim_{h \rightarrow 0} \frac{3(4 + 4h + h^2) - 4 - 2h + 1 - 9}{h} \\ &= \lim_{h \rightarrow 0} \frac{12 + 12h + 3h^2 - 4 - 2h + 1 - 9}{h} \\ &= \lim_{h \rightarrow 0} \frac{10h + 3h^2}{h} = \lim_{h \rightarrow 0} (10 + 3h) = 10 \end{aligned}$$

(Check: $f'(x) = 6x - 2$, so $f'(2) = 10$. ✓)

(b) $f(x) = 1/x$:

$$f'(x) = \lim_{h \rightarrow 0} \frac{1/(x+h) - 1/x}{h} = \lim_{h \rightarrow 0} \frac{x - (x+h)}{h x(x+h)} = \lim_{h \rightarrow 0} \frac{-1}{x(x+h)} = -\frac{1}{x^2}$$

(c) $f(x) = \sqrt{x}$, $x_0 = 4$:

$$f'(4) = \lim_{h \rightarrow 0} \frac{\sqrt{4+h} - \sqrt{4}}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{4+h} - 2}{h} \cdot \frac{\sqrt{4+h} + 2}{\sqrt{4+h} + 2} = \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{4+h} + 2)} = \frac{1}{4} = 0.25$$

Solution

A2. Limits

(a) $\frac{x^2 - 9}{x - 3} = \frac{(x-3)(x+3)}{x-3} = x+3$ for $x \neq 3$. $\lim_{x \rightarrow 3} (x+3) = 6$.

(b) Expand $(2+h)^3 = 8 + 12h + 6h^2 + h^3$. $\frac{(2+h)^3 - 8}{h} = \frac{12h + 6h^2 + h^3}{h} = 12 + 6h + h^2 \rightarrow 12$ as $h \rightarrow 0$.

(This is the derivative of x^3 at $x = 2$: $3(2^2) = 12$. ✓)

(c) Using the standard result $\lim_{u \rightarrow 0} (e^u - 1)/u = 1$: let $u = 3x$, so as $x \rightarrow 0$, $u \rightarrow 0$: $\frac{e^{3x} - 1}{x} = 3 \cdot \frac{e^{3x} - 1}{3x} \rightarrow 3 \cdot 1 = 3$.

(This is the derivative of e^{3x} at $x = 0$: $3e^0 = 3$. ✓)

(d) Divide numerator and denominator by x^2 : $\frac{4x^2 - 3x}{7x^2 + 2} = \frac{4 - 3/x}{7 + 2/x^2} \rightarrow \frac{4 - 0}{7 + 0} = \frac{4}{7}$.

Solution**A3. Continuity**

(a) $f(x) = (x^2 - 4)/(x - 2)$ is not defined at $x = 2$.

For $x \neq 2$: $f(x) = (x - 2)(x + 2)/(x - 2) = x + 2$. $\lim_{x \rightarrow 2} f(x) = 4$, but $f(2)$ is undefined.

Not continuous at $x = 2$ (removable discontinuity).

(b) Left-hand limit at $x = 1$: $\lim_{x \rightarrow 1^-} x^2 = 1$.

Right-hand limit: $\lim_{x \rightarrow 1^+} (2x - 1) = 1$.

$g(1) = 2(1) - 1 = 1$.

All three agree: g is continuous at $x = 1$.

(c) $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 = h(0)$.

h is continuous at $x = 0$ (with the given definition $h(0) = 1$).

11.2 Solutions: Section B**Solution****B1. Standard Differentiation Rules**

(a) $f'(x) = 20x^3 - 9x^2 + 4x - 7$

(b) Product rule with $u = 2x^3 - 1$, $v = x^2 + 3x$:

$$g'(x) = 6x^2(x^2 + 3x) + (2x^3 - 1)(2x + 3) = 6x^4 + 18x^3 + 4x^4 + 6x^3 - 2x - 3 = 10x^4 + 24x^3 - 2x - 3$$

(c) Quotient rule with $u = x^2 - 2x + 1 = (x - 1)^2$, $v = x^2 + 1$:

$$h'(x) = \frac{(2x - 2)(x^2 + 1) - (x^2 - 2x + 1)(2x)}{(x^2 + 1)^2} = \frac{2x^3 + 2x - 2x^2 - 2 - 2x^3 + 4x^2 - 2x}{(x^2 + 1)^2} = \frac{2x^2 - 2}{(x^2 + 1)^2}$$

(d) Chain rule: $k'(x) = e^{3x^2 - 1} \cdot 6x = 6xe^{3x^2 - 1}$

(e) $\ell(x) = \ln(x^2 + 1) - \ln(x - 1)$:

$$\ell'(x) = \frac{2x}{x^2 + 1} - \frac{1}{x - 1} = \frac{2x(x - 1) - (x^2 + 1)}{(x^2 + 1)(x - 1)} = \frac{x^2 - 2x - 1}{(x^2 + 1)(x - 1)}$$

(f) Product rule: $u = x^3$, $v = e^{-2x}$:

$$m'(x) = 3x^2e^{-2x} + x^3(-2)e^{-2x} = e^{-2x}(3x^2 - 2x^3) = x^2e^{-2x}(3 - 2x)$$

Solution**B2. Chain Rule**

(a) $y = (3x^2 + 2x - 1)^5$, inner $u = 3x^2 + 2x - 1$:

$$y' = 5(3x^2 + 2x - 1)^4 \cdot (6x + 2) = 10(3x + 1)(3x^2 + 2x - 1)^4$$

(b) $y = e^{-x^2}$, inner $u = -x^2$:

$$y' = e^{-x^2} \cdot (-2x) = -2xe^{-x^2}$$

(c) $y = \ln(\ln x)$, $x > 1$. Let $u = \ln x$:

$$y' = \frac{1}{u} \cdot \frac{1}{x} = \frac{1}{x \ln x}$$

(d) $y = \left(\frac{x+1}{x-1}\right)^3$. Let $u = \frac{x+1}{x-1}$:

$$\frac{du}{dx} = \frac{(x-1) - (x+1)}{(x-1)^2} = \frac{-2}{(x-1)^2}$$

$$y' = 3 \left(\frac{x+1}{x-1}\right)^2 \cdot \frac{-2}{(x-1)^2} = \frac{-6(x+1)^2}{(x-1)^4}$$

Solution

B3. Inverse Function Rule

$f(x) = 2x^3 + 3$. Find the point: $f(x) = 29 \Rightarrow 2x^3 = 26 \Rightarrow x = \sqrt[3]{13}$.

Wait: $2x^3 + 3 = 29 \Rightarrow x^3 = 13 \Rightarrow x = 13^{1/3}$.

$f'(x) = 6x^2$, so $f'(13^{1/3}) = 6(13^{1/3})^2 = 6 \cdot 13^{2/3}$.

Inverse function rule:

$$\left. \frac{df^{-1}}{dy} \right|_{y=29} = \frac{1}{f'(x)} \Big|_{x=13^{1/3}} = \frac{1}{6 \cdot 13^{2/3}}$$

Solution

B4. Higher-Order Derivatives

(a) $f(x) = x^5 - 4x^3 + 2x$:

$$f'(x) = 5x^4 - 12x^2 + 2, \quad f''(x) = 20x^3 - 24x, \quad f'''(x) = 60x^2 - 24$$

(b) $f(x) = xe^x$:

$$f'(x) = e^x + xe^x = (1+x)e^x$$

$$f''(x) = e^x + (1+x)e^x = (2+x)e^x$$

$$f'''(x) = e^x + (2+x)e^x = (3+x)e^x$$

Pattern: $f^{(n)}(x) = (n+x)e^x$.

(c) $f(x) = \ln x$:

$$f'(x) = x^{-1}, \quad f''(x) = -x^{-2}, \quad f'''(x) = 2x^{-3}$$

Pattern: $f^{(n)}(x) = \frac{(-1)^{n-1}(n-1)!}{x^n}$ for $n \geq 1$.

11.3 Solutions: Section C

Solution**C1. Partial Derivatives**(a) $f = x^3y^2 - 4x^2y + 3y^3$:

$$\begin{aligned}f_x &= 3x^2y^2 - 8xy \\f_y &= 2x^3y - 4x^2 + 9y^2 \\f_{xx} &= 6xy^2 - 8y \\f_{yy} &= 2x^3 + 18y \\f_{xy} &= 6x^2y - 8x = f_{yx} \checkmark\end{aligned}$$

(b) $f = e^{x^2+y^2}$:

$$\begin{aligned}f_x &= 2xe^{x^2+y^2} \\f_y &= 2ye^{x^2+y^2} \\f_{xx} &= 2e^{x^2+y^2} + 4x^2e^{x^2+y^2} = (2 + 4x^2)e^{x^2+y^2} \\f_{yy} &= (2 + 4y^2)e^{x^2+y^2} \\f_{xy} &= 2x \cdot 2ye^{x^2+y^2} = 4xye^{x^2+y^2} = f_{yx} \checkmark\end{aligned}$$

(c) $f = \ln x + \ln y + x^2/y$:

$$\begin{aligned}f_x &= \frac{1}{x} + \frac{2x}{y} \\f_y &= \frac{1}{y} - \frac{x^2}{y^2} \\f_{xx} &= -\frac{1}{x^2} + \frac{2}{y} \\f_{yy} &= -\frac{1}{y^2} + \frac{2x^2}{y^3} \\f_{xy} &= -\frac{2x}{y^2} = f_{yx} \checkmark\end{aligned}$$

Solution**C2. Total Differential**(a) $z = x^2y^3 - 3xy + 5y$:

$$dz = (2xy^3 - 3y) dx + (3x^2y^2 - 3x + 5) dy$$

(b) $Q = K^{0.4}L^{0.6}$.

$$dQ = 0.4K^{-0.6}L^{0.6} dK + 0.6K^{0.4}L^{-0.4} dL$$

At $K = 100$, $L = 64$: compute base values.

$$K^{-0.6} = 100^{-0.6} = (10^2)^{-0.6} = 10^{-1.2} \approx 0.06310$$

$$L^{0.6} = 64^{0.6} = (2^6)^{0.6} = 2^{3.6} \approx 12.126$$

$$0.4K^{-0.6}L^{0.6} = 0.4 \times 0.06310 \times 12.126 \approx 0.3061$$

$$K^{0.4} = 100^{0.4} = 10^{0.8} \approx 6.310$$

$$L^{-0.4} = 64^{-0.4} = 2^{-2.4} \approx 0.18946$$

$$0.6K^{0.4}L^{-0.4} = 0.6 \times 6.310 \times 0.18946 \approx 0.7170$$

$$\text{With } dK = 1, dL = 1: dQ \approx 0.3061(1) + 0.7170(1) = 1.023$$

So output rises by approximately **1.023 units**.

Solution

C3. Young's Theorem Verification

$$f(x, y) = x^2e^y + 3x \ln y, y > 0.$$

$$f_x = 2xe^y + 3 \ln y$$

$$f_{xy} = \frac{\partial}{\partial y}(2xe^y + 3 \ln y) = 2xe^y + \frac{3}{y}$$

$$f_y = x^2e^y + \frac{3x}{y}$$

$$f_{yx} = \frac{\partial}{\partial x}\left(x^2e^y + \frac{3x}{y}\right) = 2xe^y + \frac{3}{y}$$

$$f_{xy} = f_{yx} = 2xe^y + 3/y. \checkmark \text{ Young's Theorem verified.}$$

11.4 Solutions: Section D

Solution

D1. Implicit Differentiation

$$(a) F(x, y) = x^3 + y^3 - 6xy = 0.$$

$$F_x = 3x^2 - 6y, \quad F_y = 3y^2 - 6x.$$

$$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{3x^2 - 6y}{3y^2 - 6x} = \frac{2y - x^2}{y^2 - 2x}$$

$$(b) F(x, y) = e^{xy} - x - y^2 = 0.$$

$$F_x = ye^{xy} - 1, \quad F_y = xe^{xy} - 2y.$$

$$\frac{dy}{dx} = -\frac{ye^{xy} - 1}{xe^{xy} - 2y} = \frac{1 - ye^{xy}}{xe^{xy} - 2y}$$

$$(c) F(Q, P) = \alpha Q^2 - \beta Q + P = 0.$$

$$F_Q = 2\alpha Q - \beta, \quad F_P = 1.$$

$$\frac{dQ}{dP} = -\frac{F_P}{F_Q} = -\frac{1}{2\alpha Q - \beta}$$

For this to be well-defined, need $2\alpha Q \neq \beta$, i.e. $Q \neq \beta/(2\alpha)$. Since $D'(P) < 0$ economically, we need $2\alpha Q > \beta$, giving $dQ/dP < 0$. $\checkmark$

Solution**D2. Market Comparative Statics**(a) Equilibrium: $a - bP + cY = d + eP - fW$

$$P^* = \frac{(a - d) + cY + fW}{b + e}$$

$$Q^* = d + eP^* = d + \frac{e[(a - d) + cY + fW]}{b + e} = \frac{d(b + e) + e(a - d) + ecY + efW}{b + e} = \frac{db + ea + ecY + efW}{b + e}$$

(b) Comparative statics:

$$\frac{dP^*}{dY} = \frac{c}{b + e} > 0 \quad \frac{dP^*}{dW} = \frac{f}{b + e} > 0$$

$$\frac{dQ^*}{dY} = \frac{ec}{b + e} > 0 \quad \frac{dQ^*}{dW} = \frac{ef}{b + e} > 0$$

(c) Interpretation:

- $dP^*/dY > 0$: higher consumer income shifts demand right, raising equilibrium price.
- $dP^*/dW > 0$: higher wage raises production costs, shifting supply left, raising price.
- $dQ^*/dY > 0$: higher income raises both price and quantity (normal good).
- $dQ^*/dW > 0$: higher wages raise price enough that demand increases net — wait: if supply shifts left and demand is unchanged, Q^* should fall. Let us recheck.

Re-examination: Supply: $Q^s = d + eP - fW$. Higher W reduces supply for given P (negative supply shift). At new equilibrium: P rises and Q falls.

$$\begin{aligned} Q^* &= d + eP^* - fW = d + \frac{e[(a - d) + cY + fW]}{b + e} - fW \\ &= \frac{d(b + e) + e(a - d) + ecY + efW - f(b + e)W}{b + e} \\ &= \frac{db + ea + ecY + W[ef - fb - fe]}{b + e} \\ &= \frac{db + ea + ecY - fbW}{b + e} \end{aligned}$$

Therefore $\frac{dQ^*}{dW} = \frac{-fb}{b + e} < 0$.

The quantity falls when the wage rises (supply contraction dominates). The earlier formula was wrong — apologies.

Corrected: $\frac{dQ^*}{dW} = \frac{-fb}{b + e} < 0$: higher wages reduce equilibrium quantity.

(d) If $b + e = 0$: the denominator vanishes, so $dP^*/d\alpha \rightarrow \infty$. Economically: if the demand slope equals the negative of the supply slope ($b = -e$, but since $b, e > 0$ this would require one to be negative), the equilibrium is not well-defined or indeterminate. In standard models $b, e > 0$ so $b + e > 0$ always.

Solution**D3. Monopoly Comparative Statics**

$$(a) \pi(Q) = (A - bQ)Q - cQ^2 - dQ - F = AQ - bQ^2 - cQ^2 - dQ - F = AQ - (b + c)Q^2 - dQ - F$$

$$\text{FOC: } \pi'(Q) = A - (2b + 2c)Q - d = 0$$

$$Q^* = \frac{A - d}{2(b + c)}$$

$$P^* = A - bQ^* = A - \frac{b(A - d)}{2(b + c)} = \frac{A(b + 2c) + bd}{2(b + c)}$$

$$\text{SOC: } \pi''(Q) = -2(b + c) < 0 \checkmark \text{ (global maximum)}$$

(b) Direct differentiation:

$$\frac{dQ^*}{dA} = \frac{1}{2(b + c)} > 0 \quad (\text{higher demand intercept raises optimal output})$$

$$\frac{dQ^*}{db} = -\frac{A - d}{2(b + c)^2} < 0 \quad (\text{steeper demand reduces output})$$

$$\frac{dQ^*}{dc} = -\frac{A - d}{2(b + c)^2} < 0 \quad (\text{higher marginal cost coefficient reduces output})$$

$$\frac{dQ^*}{dd} = \frac{-1}{2(b + c)} < 0 \quad (\text{higher marginal cost reduces output})$$

$$(c) P^* = A - bQ^* = A - \frac{b(A - d)}{2(b + c)}.$$

$$\frac{dP^*}{dA} = 1 - \frac{b}{2(b + c)} = \frac{2b + 2c - b}{2(b + c)} = \frac{b + 2c}{2(b + c)}$$

$$\text{We need to show } 0 < \frac{b + 2c}{2(b + c)} < 1.$$

Lower bound: $b + 2c > 0$ and $2(b + c) > 0$, so the ratio is positive. $\checkmark$

Upper bound: $b + 2c < 2(b + c) \iff b + 2c < 2b + 2c \iff 0 < b$. $\checkmark$

Interpretation: When demand shifts up by 1 (higher A), the monopoly price rises by less than 1. The monopolist passes on only a fraction of the demand increase as higher price, partly absorbing it into higher output.

(d) By the Envelope Theorem:

$$\frac{d\pi^*}{dF} = \left. \frac{\partial \pi}{\partial F} \right|_{Q=Q^*} = -1 \quad (\text{every \$1 rise in fixed cost reduces profit by \$1})$$

$$\frac{d\pi^*}{dA} = \left. \frac{\partial \pi}{\partial A} \right|_{Q=Q^*} = Q^* = \frac{A - d}{2(b + c)} > 0$$

(Higher demand raises profit by Q^* per unit increase in A — Hotelling's Lemma for the demand parameter.)

Solution**D4. Elasticity****(a)** $Q = 200 - 4P + 0.5Y$ at $P = 10, Y = 400$:

$$Q = 200 - 40 + 200 = 360.$$

$$(i) \text{ Price elasticity: } \varepsilon_{Q,P} = \frac{dQ}{dP} \cdot \frac{P}{Q} = (-4) \cdot \frac{10}{360} = -\frac{40}{360} = -\frac{1}{9} \approx -0.111$$

$$\text{Income elasticity: } \varepsilon_{Q,Y} = \frac{dQ}{dY} \cdot \frac{Y}{Q} = 0.5 \cdot \frac{400}{360} = \frac{200}{360} = \frac{5}{9} \approx 0.556$$

(ii) $|\varepsilon_{Q,P}| = 1/9 < 1$: demand is **inelastic** at this point.**(b)** $TR = PQ$. Differentiate with respect to Q :

$$MR = \frac{d(PQ)}{dQ} = P + Q \frac{dP}{dQ} = P \left(1 + \frac{Q}{P} \frac{dP}{dQ} \right) = P \left(1 + \frac{1}{\varepsilon_{Q,P}} \right)$$

(using $1/\varepsilon_{Q,P} = \frac{P}{Q} \cdot \frac{dQ}{dP}$ so $\frac{Q}{P} \cdot \frac{dP}{dQ} = \frac{1}{\varepsilon_{Q,P}}$.) ✓

Implications: If $|\varepsilon| > 1$ (elastic), $\varepsilon < -1$, so $1 + 1/\varepsilon \in (0, 1)$: $MR > 0$. If $|\varepsilon| < 1$ (inelastic), $1 + 1/\varepsilon < 0$: $MR < 0$. Profit-maximising firms never operate in the inelastic range (since MR must equal $MC > 0$).

(c) $Q = AK^\alpha L^{1-\alpha}$:

$$\varepsilon_{Q,K} = \frac{\partial Q}{\partial K} \cdot \frac{K}{Q} = \alpha AK^{\alpha-1} L^{1-\alpha} \cdot \frac{K}{AK^\alpha L^{1-\alpha}} = \alpha \quad \checkmark$$

$$\varepsilon_{Q,L} = \frac{\partial Q}{\partial L} \cdot \frac{L}{Q} = (1-\alpha) AK^\alpha L^{-\alpha} \cdot \frac{L}{AK^\alpha L^{1-\alpha}} = 1-\alpha \quad \checkmark$$

Euler's Theorem: F is homogeneous of degree $\alpha + (1-\alpha) = 1$, so:

$$K \cdot \frac{\partial Q}{\partial K} + L \cdot \frac{\partial Q}{\partial L} = K \cdot \frac{\alpha Q}{K} + L \cdot \frac{(1-\alpha)Q}{L} = \alpha Q + (1-\alpha)Q = Q \cdot 1 = Q$$

This is Euler's Theorem for degree 1 (CRS): the sum of input payments at marginal products exhausts output (zero-profit condition). ✓

Summary Table

Concept	Key Formula / Statement
Difference quotient	$\Delta y / \Delta x = [f(x + \Delta x) - f(x)] / \Delta x$
Derivative	$f'(x) = \lim_{h \rightarrow 0} [f(x + h) - f(x)] / h$
Product rule	$(fg)' = f'g + fg'$
Quotient rule	$(f/g)' = (f'g - fg') / g^2$
Chain rule	$dy/dx = (dy/du)(du/dx)$
Inverse function	$dx/dy = 1/(dy/dx)$
Impl. function rule	$dy/dx = -F_x/F_y$ (from $F(y, x) = 0$)
Total differential	$dz = f_x dx + f_y dy$
Young's Theorem	$f_{xy} = f_{yx}$ (if both continuous)
Comparative statics	$dy^*/d\alpha$ from differentiating equilibrium condition
Envelope Theorem	$dV/d\alpha = \partial f(x^*; \alpha) / \partial \alpha$
Concavity (1D)	$f'' < 0$; local max $\Rightarrow$ global max if on convex domain
Point elasticity	$\varepsilon_{y,x} = (dy/dx)(x/y) = d \ln y / d \ln x$
MR-Elasticity	$MR = P(1 + 1/\varepsilon_{Q,P})$
Homogeneity deg. k	$f(tx) = t^k f(x)$; Euler: $\sum x_i f_{x_i} = kf$

References

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End of Notes — ECON 320 Comparative Statics & Derivatives — May 2026